

The Nominal Left Periphery in German: Abstract Agreement Chains

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As is well known, concord holds between different nominal elements of the noun phrase. This paper identifies a restriction on this process. Given certain grammaticality contrasts, this restriction becomes apparent in the DP-layer if an element occurs in the left periphery, that is, above the DP-layer. I propose that an abstract agreement chain is built in syntax that must exclude (possessive) genitive features. It will become clear that the DP-layer plays an important role. First, this restriction only manifests itself in the DP-layer. Second, the DP-layer of definite DPs must contain an overt element – under certain conditions, this layer attracts items like *alle* ‘all’ and *beide* ‘both’, which can be merged in different positions. Briefly extending the discussion to other Germanic languages, it appears that only languages with genitive case exhibit this restriction on the agreement chain.

1. Introduction

It is well known that elements in the noun phrase show concord; that is, they share grammatical features like case, number, and gender:¹

- (1) a. de-r klein-e nett-e Mann
the-NOM.MASC short-NOM.MASC nice-NOM.MASC man.NOM.MASC
‘the short nice man’
b. mit all-en mein-en nett-en Freunde-n
with all-DAT.PL my-DAT.PL nice-DAT.PL friends-DAT.PL
‘with all my nice friends’

With this in mind, Bayer (2015) discusses the following grammaticality contrast in German where inflected *alle* ‘all’ can precede a possessive determiner (2a) or a Possessor Doubling construction (2b) but not a Saxon Genitive (2c):

- (2) a. all-e ihr-e Bücher
all-INFL her-INFL books
‘all her books’
b. all-e de-r Maria ihr-e Bücher
all-INFL the-DAT Mary her-INFL books
‘all Mary’s books’

¹ In the gloss, I only provide details necessary for understanding the main points of the paper. As such, moving forward, case within the noun phrase is usually only indicated if it diverges from the concord case feature of the larger noun phrase. Also, inflections showing concord features are simply indicated by INFL. Abbreviations are as follows: NOM = nominative, ACC = accusative, DAT = dative, GEN = genitive; MASC = masculine, NEUT = neuter, FEM = feminine; PL = plural; REFL = reflexive pronoun.

- c. * all-e Marias Bücher
all-INFL Mary's books
'all Mary's books'

Assuming that the dative phrase *der Maria* '(the) Mary' is in Spec,DP, Bayer argues that this yields an uninterrupted agreement chain such that all relevant elements participate in concord of the noun phrase. This is indicated by +AGR in (3a). In contrast, the chain in (3b) is disrupted by *Marias* 'Mary's', which does not share features with the other elements:

- (3) a. [XP all-e [DP (der Maria) [D' ihr-e [NP Büch-er]]]]
+AGR +AGR +AGR
b. [XP all-e [DP Maria [D' 's [NP Büch-er]]]]
+AGR -AGR +AGR

Bayer (2015) proposes that the disruption of the agreement chain in (3b) leads to ungrammaticality. He tentatively suggests that cases involving *beide* 'both' also follow from this account (4a). As noted by Bayer, the Saxon Genitive can precede the quantifier *beide* in (4b) but not *alle* in (4c):

- (4) a. * beid-e Marias Bücher
both-INFL Mary's books
b. Marias beid-e Bücher
Mary's both-INFL books
'both Mary's books'
c. * Marias all-e Bücher
Mary's all-INFL books

In this paper, I argue that Bayer (2015) is basically right in that an agreement chain is formed in (2). However, I demonstrate that this chain is independent of the overt inflection on *alle* 'all' – uninflected *all* and preposed PPs also lead to ungrammaticality in this context. I argue that this phenomenon is due to a selectional restriction. The latter is imposed on the DP-layer by an element in the left periphery, here defined as the structural level above the DP-layer. Regular concord builds an abstract agreement chain in syntax. However, given the requirements of the higher selecting head, that agreement chain must exclude a (possessive) genitive feature in the DP-layer. If no element is present in the left periphery, no such restriction on the agreement chain holds. To be clear, I propose that there is an additional requirement for concord if the left periphery is projected. As for (4a), it will be argued that this data point does not fall under this proposal but has a different, independent explanation – *beide* 'both' cannot be merged in the left periphery in the first place.

The paper is organized as follows. In Section 2, I provide more data and discuss some previous proposals. Section 3 offers my proposal about abstract agreement chains and discusses some initial consequences, extensions, and predictions. In Section 4, I turn to some further extensions, and Section 5 closes the paper.

2. Data and Previous Proposals

To begin, I discuss more relevant data showing that elements in the left periphery do not have to agree in concord features to trigger this restriction and that agreement chains do not require overt inflections on elements in the DP-layer. Then, it is made explicit that elements below the DP-layer are not subject to the same restriction. Finally, I briefly discuss some previous work.

2.1. More Data

Based on similar data in Haider (1988, 1992), Bayer (2015) discusses the following grammaticality contrast in German where inflected *alle* ‘all’ can precede a possessive determiner (5a) or a Possessor Doubling construction (5b) but not a Saxon Genitive (5c):²

- (5) a. all-e ihr-e Bücher
all-INFL her-INFL books
‘all her books’
b. all-e de-r Maria ihr-e Bücher
all-INFL the-DAT Mary her-INFL books
‘all Mary’s books’
c. * all-e Marias Bücher
all-INFL Mary’s books
‘all Mary’s books’

It is interesting to point out that all examples in Bayer (2015) have agreement inflections on the determiner(-like) elements. However, German is ungrammatical even when no agreement inflection is on the element in the left periphery. Comparing (5) to (6), it should be observed that (6b) is worse than (5b) (cf. Haider 1988: 53, Karnowski & Pafel 2004: 182 fn. 13; also Haider 1992: 314-15, who judges (6b) to be fully ungrammatical):³

- (6) a. all ihr-e Bücher
all her-INFL books
‘all her books’
b. ?? all de-r Maria ihr-e Bücher
all the-DAT Mary her-INFL books
‘all Mary’s books’
c. * all Marias Bücher
all Mary’s books
‘all Mary’s books’

² The example in (5b) is provided with Bayer’s judgments. There are two differences for me: While for Bayer, the proprial article *der* is obligatory, it is not in my (northern) dialect (for the discussion of regional differences in the usage of the proprial article in German, see Nübling *et al* 2015: 123-28). In fact, its presence leads to degraded grammaticality for me here. Having said that, even in the absence of *der*, the example is still slightly marked for me. I return to this degradedness in Section 4.2.

³ Strictly speaking, not all of Bayer (2015)’s examples have overt inflections on the determiner(-like) elements. In his footnote 1, Bayer provides (6c). He states that the absence of the inflection *-e* on *alle* is mainly due to phonological reasons. This is consistent with the fact that he argues throughout the paper that German has prenominal agreement.

Considering *all* vs. *alle*, the agreement chain proposed by Bayer cannot be about overt inflection in the left periphery as the same facts hold for uninflected *all*.⁴ The question arises if this agreement chain involves some kind of abstract agreement.

As it turns out, this agreement chain cannot be about agreement in the left periphery at all. There is a construction not discussed by Bayer (2015) that involves a preposed PP (7a). Importantly, PPs do not share agreement features with the rest of the nominal. Furthermore, note already here that there is also no overt inflection on *sein* ‘his’ in (7b). Independently of these two facts, the same contrast as discussed above holds such that a possessive determiner is fine (7b) but a Saxon Genitive is not (7c) (data are taken from Haider 1992: 315; for (7c), see also Fortmann 1996: 22):

- (7) a. aus Jena de-r Anruf
from Jena the-INFL call
‘the call from Jena’
b. aus Jena sein Anruf
from Jena his call
‘his call from Jena’
c. * aus Jena Peters Anruf
from Jena Peter’s call

To reiterate, it is clear that there is no disruption of an agreement chain by *Peters* in (7c) as the preposed PP does not participate in agreement in the first place. Still, the example is ungrammatical. This means that the notion of agreement chain has to be refined.

As just discussed, elements in the left periphery do not have to be inflected (and do not have to agree in concord features). Indeed, as already documented in (7b) above, the agreement chain cannot be about overt inflection on an element in the DP-layer either. This extends to other elements that may occur without overt inflection, the indefinite article *ein* ‘a’ provided below as *’n* and the singularity numeral *ein* ‘one’ provided below as *EIN* (the singularity numeral is typically stressed and is put in capital letters):

- (8) a. von Peter ’n Freund
of Peter a friend
‘a friend of Peter’s’
b. (?) von Peter EIN Freund
of Peter one friend
‘one friend of Peter’s’

In other words, the absence of agreement inflection here does not necessarily lead to degradedness either. To be clear then, while the Saxon Genitive *Marias* leads to ungrammaticality, the uninflected possessive determiner *sein* (7b), the indefinite article *’n* (8a),

⁴ Note that if this were about an overt agreement chain, that requirement would have to be met before the inflection on *alle* ‘all’ was deleted (cf. Footnote 3). As we will see momentarily though, this chain cannot be about inflections, overt or deleted, at all.

or the numeral *EIN* (8b) do not. Importantly, the latter three elements have the same distribution as their inflected plural counterparts *seine*, *'ne*, and *EINE*.⁵

- (9) a. von Peter sein-e Autos
from Peter his-INFL cars
'his cars from Peter'
b. von Peter 'n-e Freundin
of Peter a-INFL girlfriend
'a girlfriend of Peter's'
c. (?) von Peter EIN-E Freundin
of Peter one-INFL girlfriend
'one girlfriend of Peter's'

Focusing on nominative case, the complete set of these three elements in the masculine, neuter, feminine, and plural is given in (10). Note that in ellipsis contexts, the masculine and neuter forms have an inflection provided in parentheses in (10a-b) (for recent discussion, see Murphy 2018). In the feminine and plural, these inflections are obligatory, independent of ellipsis (10c-d):⁶

- (10) a. masculine: sein(-er), ein(-er), EIN(-ER)
his-INFL, an-INFL, one-INFL
b. neuter: sein(-es), ein(-es), EIN(-ES)
his-INFL, an-INFL, one-INFL
c. feminine: sein-e, ein-e, EIN-E
his-INFL, an-INFL, one-INFL
d. plural: sein-e, ein-e
his-INFL, some-INFL

Given this paradigm, it is clear that the appropriately inflected form must be inserted in a given morpho-syntactic context. In other words, these elements do agree with their head noun albeit not overtly in every case.

To sum up, Saxon Genitives cannot be present if the left periphery is occupied by an element. This is independent of whether or not that element agrees with the lower part of the nominal. Second, the determiner(-like) element in the DP-layer does not have to show overt agreement inflection to yield a grammatical example.

⁵ In the presence of a possessive determiner like *sein* 'his', a preposed PP as in (9a) must have a non-possessive interpretation (Roehrs 2020).

⁶ Traditionally, these elements are called *ein*-words – they all involve the form *ein*. To make their relatedness clearer, I provide the indefinite article here as the unreduced form *ein*, rather than *'n*. Also, note that plural *eine* occurs only in very colloquial, emotive contexts in certain dialects of German (Roehrs to appear):

(i) *Was für 'n-e / Was für ein-e Idioten!* (Northern German)
what for.a-INFL / what for a-INFL idiots
'What kind of idiots!'

Discussing Dutch, Bennis *et al.* (1998) call this element spurious article. I translate German *eine* in (10d) as 'some'.

2.2. Elements Below the DP-Layer

The restriction above seems to involve elements in the DP-layer and above. Elements below these structural levels do not seem to be subject to the same restriction. While numerals and adjectives do not trigger the chains, numerals, to the extent possible, and adjectives more generally must also agree in concord features with the lower head noun.⁷ This is illustrated with numerals in (11), adjectives in (12), and noun phrases without such modifiers in (13). Importantly for current purposes, observe that all these cases are fine when occurring with or without a preceding determiner or determiner-like element (ST = strong ending, WK = weak ending):

- (11) a. di-e /dies-e /mein-e zehn Bücher
the-INFL/these-INFL/my-INFL ten books
'the/these/my ten books'
- b. Peters zehn Bücher
Peter's ten books
'Peter's ten books'
- c. zehn Bücher
ten books
'ten books'
- (12) a. di-e /dies-e /mein-e gut-en Bücher
the-INFL/these-INFL/my-INFL good-INFL.WK books
'the/these/my good books'
- b. Peters gut-e Bücher
Peter's good-INFL.ST books
'Peter's good books'
- c. gut-e Bücher
good-INFL.ST books
'good books'
- (13) a. di-e /dies-e /mein-e Bücher
the-INFL/these-INFL/my-INFL books
'the/these/my books'
- b. Peters Bücher
Peter's books
'Peter's ten books'
- c. Bücher
books
'books'

⁷ As discussed above, the singularity numeral *EIN* 'one' exhibits agreement inflection in most morpho-syntactic contexts. In contrast, *zwei* 'two' and *drei* 'three' only show agreement inflection in the genitive plural (e.g., *zwei-er*), but numerals above that do not show any agreement inflection.

Note that the inflection on the adjective in (12) alternates between a weak ending in (12a) and a strong one in (12b-c).⁸ Notice that this inflectional alternation is independent of the chain discussed above, which has to do with agreement (determiners) vs. non-agreement (Saxon Genitives).

Having said that, there is one well-known restriction on adjectives. However, it is of a different kind than that, which was discussed in Section 2.1. This restriction is often referred to as Head-Final Filter (e.g., Williams 1982). This can be illustrated for German as follows. Argument PPs can precede or follow their adjectival heads when the latter are in predicative position (14a-b). When prenominal, adjectives must be inflected taking their PP arguments on the left (14c-d) (also Haider 2022; for similar facts in Dutch, see van Riemsdijk 1998a: 673-74):

- (14) a. Der Vater ist auf seinen Sohn stolz.
the father is of his son proud
'The father is proud of his son.'
- b. Der Vater ist stolz auf seinen Sohn.
the father is proud of his son
'The father is proud of his son.'
- c. der [auf seinen Sohn] stolz-e Vater
the of his son proud-INFL father
'the father proud of his son'
- d. * der stolz-e [auf seinen Sohn] Vater
the proud-INFL of his son father

There are more cases exhibiting this type of restriction.

As is well known, the degree element *genug* 'enough' is special in that it follows its modifiee (15a). Similar to above, the inflected adjective cannot be separated from the head noun (15b). Interestingly, when the agreement inflection appears on the degree word, the example improves for many speakers (15c). This is indicated below by a percentage sign. As might be expected, regular degree words precede inflected adjectives without a problem (15d):

- (15) a. Das Auto ist groß genug.
the car is big enough
'the car is big enough'
- b. * das groß-e [genug] Auto
the big-INFL enough car

⁸ Traditionally, it is claimed that the inflection on the adjective is determined by the inflection on the preceding determiner(-like) element: weak inflections on adjectives are brought about by inflected determiners; strong inflections on adjectives involve the elsewhere case. While true for most cases, this is not quite accurate – exceptions go in both directions (for detailed discussion, see Roehrs 2015, to appear):

- (i) a. wir nett-en Studenten
we nice-WK students
'us nice students'
- b. dies-es mein groß-es Glück
this-INFL my great-ST happiness
'this my great happiness'

In other words, the inflections on the adjectives are independent of the inflections on the determiner(-like) elements.

- c. % das groß [genug-e] Auto
the big enough-INFL car
'the big enough car'
- d. das [ausreichend] groß-e Auto
the sufficiently big-INFL car
'the sufficiently big car'

Similarly, comparatives reveal the same type of restriction: the standard of comparison (introduced by *als* 'than') also follows the comparative adjective (16a), but it cannot separate the inflected adjective from the head noun (16b-c):

- (16)
- a. Dieses Auto ist größer als deins.
this car is bigger than yours
'This car is bigger than yours.'
 - b. ein größer-es Auto [als deins]
a bigger-INFL car than yours
'a bigger car than yours'
 - c. * ein größer-es [als deins] Auto
a bigger-INFL than yours car

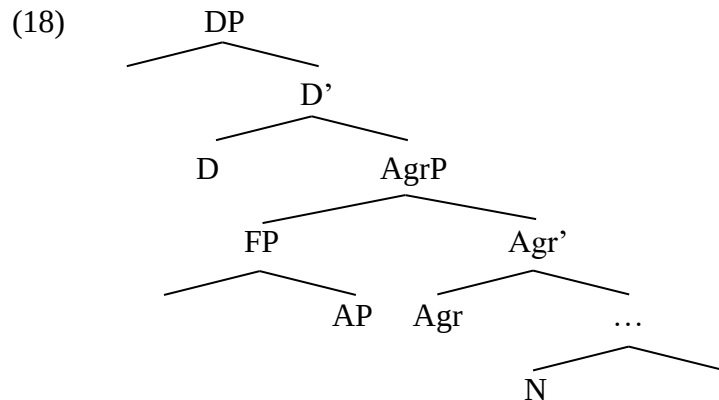
Note that the PPs and degree elements above are dependents of the adjective (not the head noun). Similarly, the standard of comparison is a dependent of the comparative adjective. In other words, the presence of the former elements depends on the presence of the latter ones – the (comparative) adjectives.

Finally, as mentioned above, adjectives are inflected in German when they occur in prenominal position. There are only a few lexical exceptions (17a) (note that *-n-* is an epenthetic consonant). Importantly, dependents on those adjectives do not require the inflections to be present (17b-c) (note that the latter two examples, retrieved by a *google* search, show *lila* spelled with a capital letter):

- (17)
- a. ein lila(-n-es) Kleid
a purple(-n-INFL) dress
'a purple dress'
 - b. ein total Lila Kinderzimmer
a totally purple children.room
'a totally purple children's room'
 - c. ein total Lila Bild
a totally purple picture
'a totally purple picture'

To be clear, inflections on adjectives are independent of the presence of their dependents.

Corver (1997) argues that adjectives project a complex structure in that the degree words and the arguments of the adjectives are part of that structure. In other words, adjectives involve extended projections similar to nouns (Grimshaw 1991, van Riemsdijk 1998b). Simplifying here, this is indicated by a Functional Phrase (FP) on top of AP in (18). This extended projection FP resides in the specifier position of the extended projection line of the noun (Cinque 1994, 2010):



The elements in the left periphery, preposed PPs and *all(e)*, are different. PPs can be in postnominal position where they are separated from the determiner element. In fact, postnominal PPs can occur without an (overt) determiner (e.g., *Bücher von Peter* ‘books of Peter’). As to the quantifier, *all(e)* can be separated from the determiner element by a dative noun phrase as already shown above (e.g., *alle der Maria ihre Bücher* ‘all Mary’s books’) or by another determiner element (e.g., *alle diese meine Bücher* ‘all these books of mine’; see also Section 3.3). This means that these elements are not dependent on the following determiner(-like) element and do not form an extended projection with the determiner(-like) element.

To conclude, the Head-Final Filter is a different restriction – it has to do with the ordering of dependents and their heads in prenominal position, but not with the (non-)agreement of the elements involved. I turn to some previous proposals of the restriction documented in Section 2.1.

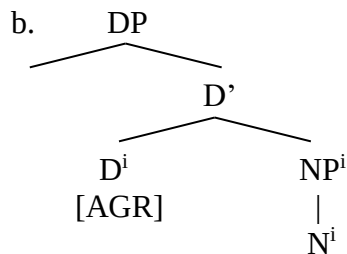
2.3. Previous Proposals

While Haider (1988) seems to be the first to have observed this grammaticality contrast, he does not discuss it in much detail. Haider proposes that *all* is in Spec,DP. Assuming that the possessive determiner is in D but that the Saxon Genitive is in Spec,DP, this immediately explains why Saxon Genitives cannot co-occur with *all* – both compete for the same specifier position. There are two issues with this simple account. First, I will argue in the next section that preposed PPs are in a position higher than Saxon Genitives. This means that the ungrammatical cases involving Saxon Genitives and preposed PPs cannot be explained by complementary distribution. Second, there is evidence that *all* is actually in a higher position too (e.g., *all[e] diese meine Freunde* ‘all these friends of mine’, see Section 3.3).

Bayer (2015) formulates the notion of the disruption of an agreement chain. The idea of an agreement chain is based on work by Olsen (1989a). In order to explain concord inside the DP, Olsen (1989a: 45) proposes that an uninterrupted chain of identical superscripts is established (19a). Couched in current terminology, D functionally selects NP such that a chain is established going from D to N (19b):⁹

⁹ Haider (1988: 52) suggests that functional heads select features for which they themselves are specified. Note also that Olsen treats adjectival quantifiers and adjectives as adjuncts to NP.

- (19) a. *Agreement Chain*
 An agreement chain is an uninterrupted chain of identical superscripts headed by an AGR-element and formed on the basis of functional selection between AGR and its functional complement.



Given selection, concord involves a top-down relation in this type of analysis. Furthermore, the agreement chain is made visible by overt concord on all lexical heads (if possible). While Olsen does not discuss the relevant grammaticality contrast, Bayer does.

Following Olsen's idea of an agreement chain, Bayer (2015) provides an account of the restriction under discussion. Rather than superscripts, he employs the marking AGR, with a positive value if agreement holds (20a) and a negative value if it does not (20b):

- (20) a. [XP all-e [DP (der Maria) [D' ihr-e [NP Büch-er]]]]
 +AGR +AGR +AGR
 b. [XP all-e [DP Maria [D' 's [NP Büch-er]]]]
 +AGR -AGR +AGR

A negative value leads to the disruption of the agreement chain, and this is taken to explain the ungrammaticality. Note though that the status of AGR itself is not made explicit. In other words, it is not entirely clear in what sense this involves a chain. Furthermore, as demonstrated above, the agreement chain postulated by Bayer (2015) cannot be about agreement with the leftmost element – preposed PPs like *von Peter* ‘of Peter’ may also lead to ungrammaticality – and it cannot be about overt inflection on the following determiner – uninflected elements like *sein* ‘his’ and *ein* ‘a’ do not bring about ungrammaticality.

Cirillo (2016) takes a cross-linguistic view of this type of data discussing German, Dutch, and English (for the discussion of languages other than German, see Section 3.4). To account for the German data, Cirillo makes use of the following main components: a definite DP must be occupied by an overt element in either D or Spec,DP (also Longobardi 1994, Julien 2005), Saxon Genitives involve genitives where -s is a case ending assigned in Spec,NP, and the head noun undergoes partial raising to the head n of a nP in a Split-NP analysis. To provide some initial illustration, if an overt element is in the DP-layer, then the Saxon Genitive stays in situ deriving postnominal possessives (21a). If there is no overt element in the DP-layer, then the Saxon Genitive moves to Spec,DP licensing the definite DP in the process (21b):

- (21) a. [DP die [n_P Bücher_i [n_P Marias t_i]]]
 b. [DP Marias_k [n_P Bücher_i [n_P t_k t_i]]]

Turning to the restriction discussed above, Cirillo assumes that inflected *alle* ‘all’ is a complex element that consists of two linguistic items in different phrasal levels: on the one hand, the quantifier *all* selects a definite DP; on the other, the ending *-e* is a determiner-like element cautiously interpreted as the phonetic realization of the definiteness feature in D. With this in mind, we add the bipartite quantifier to the derivation in (21) above. Now, if the Saxon Genitive stays in situ, grammatical (22a) is derived. However, if it moves to Spec,DP, ungrammatical (22b) is brought about:

- (22) a. [QP all [DP -e [nP Bücher_i [NP Marias t_i]]]]
 b. * [QP all [DP Marias_k -e [nP Bücher_i [NP t_k t_i]]]]

Recall that Saxon Genitives move to Spec,DP to license the definite DP if the DP-level is empty. Unlike in (21b) above, this is not the case in (22b) where *-e* occupies D. In other words, the reason why (22b) results in a bad derivation is that there is no motivation for the Saxon Genitive to move to the DP-level. These assumptions explain the ungrammaticality of Saxon Genitives immediately following inflected *alle*.

Cirillo (2016: 206) does not provide an analysis of uninflected *all*. Given his assumptions, it is not clear though how to rule out uninflected *all* as *-e* is missing – the Saxon Genitive should be able to move to Spec,DP to license the definite DP (23a). Furthermore, like *all*, preposed PPs, not discussed by Cirillo, are also above the DP-level and do not involve an element in D. Again, the Saxon Genitive should be able to move to Spec,DP to license the definite DP (23b):

- (23) a. * [QP all [DP Marias_k [nP Bücher_i [NP t_k t_i]]]]
 b. * [QP von Peter [DP Marias_k [nP Bücher_i [NP t_k t_i]]]]

Observe now that although the definite DP is licensed by felicitous movement of the Saxon Genitive in (23a-b), the examples are ungrammatical. It is not clear what explains that on Cirillo’s assumptions.

The current discussion has highlighted the fact that the left periphery imposes the restriction but is not part of it. It seems that this restriction involves a top-down dependency between the left periphery and the DP-layer – if the left periphery is absent, no restriction manifests itself. Like Olsen, I will employ selection to capture this (cf. also Cirillo 2016). I will formalize it as an additional requirement of the more general property of concord holding between all nominal elements that are part of the extended projection line of the noun.¹⁰ Nevertheless, following Bayer, I will often speak of the grammaticality contrast above as being due to the disruption of an agreement chain.

¹⁰ There is debate of how to capture concord (e.g., see Sigurðsson 1989:112-113 for an account based on feature percolation/spreading, Schoorlemmer 2009 for an Agree-based proposal, and Norris 2014 for an analysis in terms of feature spreading and local feature copying). Interestingly, Carnie (2021: 293) and Olsen (1989a: 45) use selection to account for concord. Note that this is the same process that I will argue below to hold between a head in the left periphery and its DP complement. If it turns out that both of these processes are indeed related, then the current account proposes an additional requirement of this selectional process in the left part of the noun phrase.

3. New Proposal

In this section, I provide my own proposal. After laying out the basic idea, I discuss some refinements, consequences, extensions, and predictions.

3.1. Basic Proposal

Preposed PPs and Saxon Genitives are possessives that do not agree with the head noun in phi-features and case. In other words, they do not participate in concord within the larger noun phrase. Having said that, there are also some important differences between them. As discussed in detail in Roehrs (2020), preposed PPs require the presence of an article but Saxon Genitives do not allow it:

- (24) a. von Peter *(das) Buch
of Peter the book
'the book of Peter's'
b. Peters (*das) Buch
Peter's the book
'Peter's book'

In fact, preposed PPs tolerate another element (*all*) to intervene, again unlike Saxon Genitives:

- (25) a. (?) von Peter all di-e Bücher
of Peter all the-INFL books
'all the books of Peter's'
b. * Peters all(-e) Bücher
Peter's all-INFL books

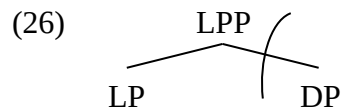
Besides these structural distinctions, there are some other differences discussed in the above-mentioned article.

Preposed PPs seem to be sensitive to information structure in a way Saxon Genitives are not. Specifically, *Peter* in *von Peter* in (24a) has (contrastive) stress on it and must have been mentioned before and/or is contrasted with another person. This is different for *Peter* in *Peters* in (24b), where the Saxon Genitive can appear in out-of-the-blue contexts. In addition, Saxon Genitives can also have the features and functions just mentioned for the preposed PPs.

In order to account for these cases, Roehrs (2020) proposes that the noun phrase is more complex, similar to the split CP (Rizzi 1997). Specifically, Roehrs argues that there is a phrase level above the DP-layer. Following Giusti & Iovino (2016), he labels this level Left Periphery Phrase (LPP). If these information structural considerations are on the right track, then this provides additional evidence that preposed PPs and Saxon Genitives are in different structural positions.

To be clear then, given these structural and information structural differences, I argue that preposed PPs are above the DP-layer, but Saxon Genitives are determiner-like elements inside the DP-level. In other words, preposed PPs are in – what I have called – the left periphery. With

this in mind, I propose that there are two domains in the noun phrase relevant to this phenomenon, the DP proper and the left periphery:



Following standard assumptions, the DP-layer is the level hosting determiners and determiner-like elements: definite and indefinite articles are in D but demonstratives, possessive determiners, and Saxon Genitives are in Spec,DP.¹¹ Linguistic units preceding those elements are taken to be in LPP making up the left periphery (e.g., PPs and certain cases of *all[e]*). It will be shown that the DP-level plays an important part in explaining the phenomenon under discussion.

Recalling the data involving preposed PPs, I propose *pace* Bayer (2015) that an abstract chain does not (necessarily) start in the left periphery but may start lower in the structure. This can be formulated as a generalization holding over the left edge of the noun phrase, that is, over the left periphery and the DP-layer:¹²

(27) *Generalization of the Left Edge:*

If a linguistic unit is in the left periphery, the determiner(-like) element must agree with the head noun.

Note that this formulation does not exclude cases where the agreement chain does start in the left periphery (e.g., *alle ihre Bücher* ‘all her books’). To be clear then, if there is an element above the DP-layer, then the element inside the DP-layer must agree with the head noun in phi-features and case. For expository purposes, we can state that an element in the left periphery “activates” or “triggers” the formation of an agreement chain inside the DP and that it can (but does not have to) participate in it. Given the latter optionality, I will take the element in the DP-layer as the starting point of the chain; the end point is the head noun. I will call this an “agreeing” DP. I will refer to this starting point as the HEAD of the chain, and I will refer to the element that activates the chain in the left periphery as the TRIGGER.

The trigger of the chain might be interpreted as follows. Considering that the head of the chain is below the trigger, I will claim that the head LP in (26) selects an “agreeing” DP in a sense to be made more precise below. LP is the head of LPP in the specifier of which sits the trigger. To make the account more formal, I will make use of well-established mechanisms like selection and Spec-head agreement.

Starting with the latter, Spec-head agreement is well known from Definiteness Spread where the interpretation of a possessive DP depends on the definiteness of the possessor itself:

- (28)
- a. * There was John’s book on the table.
 - b. There was a man’s dog in the garden.

¹¹ With some elements in D and others in Spec,DP, their complementary distribution cannot be a simple structural account (see Roehrs 2019).

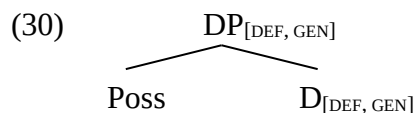
¹² Alternatively, we might state at first glance that non-agreeing elements like preposed PPs and Saxon Genitives can only occur in the left edge of the noun phrase (but not below it). We will see below though that genitives can occur in lower prenominal positions in older varieties of German.

According to Alexiadou (2005), this straightforwardly holds in English (but is different in Hebrew and Greek). German is like English. Assuming the possessor to be in Spec,DP, Spec-head agreement makes D and thus DP as a whole definite. What has not been discussed in much detail is what happens to the case feature on the element in Spec,DP.

Traditionally, possessives are taken to be in the genitive. In German, prenominal possessives can typically not be complex – the presence of a proprial article leads to markedness (29a). With case usually marked on the determiner in German, it is interesting to note though that the genitive form of the article fares best.¹³ Furthermore, there are some isolated examples where complex possessives are not impossible (29b), including some fossilized phrases (29c).¹⁴ These cases clearly show genitive:

- (29) a. ((?de-s / ?*de-m / *de-n / *de-r) Peters Auto
the-GEN / the-DAT / the-ACC / the-NOM Peter's car
'Peter's car'
- b. ?? mein-es Vaters Auto
my-GEN father's car
'my father's car'
- c. de-s Kaiser-s neue Kleider
the-GEN emperor-GEN new clothes
'the emperor's new clothes'

Both definiteness and case are typically interpreted as features. Thus, if definiteness undergoes Spec-head agreement, so should genitive. In other words, D (and thus DP) agrees with the possessive in definiteness and case. Consequently, there are at least two features on DP, a definiteness and a genitive feature:¹⁵



Thus far, Spec-head agreement as regards the (possessive) genitive and its consequences have not received much attention in the literature. This is presumably so because any effects are absent in the transition from the nominal to a higher domain; that is, when a DP is selected by a verb, a preposition, or an adjective (I turn to cases where a DP is selected by a noun below):

¹³ It is a controversial issue as to whether or not -s/'s indicates genitive or a possessive relation. For instance, it is often claimed that -s is genitive in German (Cirillo 2016: 193ff) but that 's is not in English (Alexiadou 2005: 794ff).

¹⁴ Note that more recent forms like *mein Vaters Auto* are basically perfect (for discussion, see see Fuß 2011).

¹⁵ There is also debate on where genitive is assigned/checked in the noun phrase. Abney (1987), Haider (1988, 1992), and Olsen (1989b) claim it is in Spec,DP; Alexiadou (2005: 791) and Cirillo (2016: 188) argue it is below the DP-level. For the following account to go through, the genitive feature must appear on DP (which is brought about by Spec-head agreement if an element is in Spec,DP, independently of whether it is base-generated or moved there). Furthermore, there are also regular concord features including case in the noun phrase (see (1) in the introduction). For some reason, the (concord) case feature does not clash with the (possessive) case feature in cases like *Peters Auto* 'Peter's car'. One could suggest that these genitives are two different types (and/or that they are part of two different types of feature bundles) that do not interact with one another.

- (31) a. Ich bestaune Peters Auto.
I admire Peter's car
'I admire Peter's car.'
- b. in Peters Auto
in Peter's car
'in Peter's car'
- c. Sie ist Peters Mutter ähnlich.
she is Peter's mother similar
'She is similar to Peter's mother.'

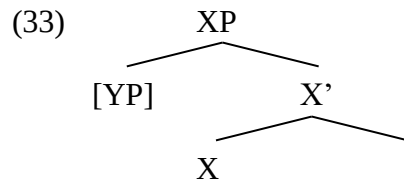
However, effects become apparent inside the projection line of the head noun itself, specifically in the transition from the DP-layer to the left periphery. Here, the effects become visible through the data discussed in Section 2.1.

Recall that I argued in Section 2.2 that degree words and PP arguments are part of the extended projection of adjectives. If the quantifier *alle* is added in the left periphery of a noun phrase with a complex adjective modifier, the example remains grammatical (32a). Similarly, as discussed in the introduction, the possessor of the Possessor Doubling construction does not disrupt the agreement chain either (32b), and certain predeterminer elements show the same behavior (32c) – it is as if these elements, indicated below by square brackets, are invisible to the formation of a chain:¹⁶

- (32) a. all-e di-e [auf de-n Sohn] stolz-en Väter
all-INFL the-INFL of the-ACC son proud-INFL fathers
'all the fathers proud of their sons'
- b. all-e [(de-m) Peter] sein-e Bücher
all-INFL the-DAT Peter his-INFL books
'all Peter's books'
- c. von Peter [so manch] ein Freund
of Peter so some a friend
'some friends of Peter's'

Similar to degree words and PP arguments of adjectives, the dative possessor and the complex predeterminer depend on the presence of the following element. In other words, the possessor cannot occur without the possessive determiner (32b), and the complex predeterminer requires the presence of the indefinite article (32c). The reverse does not hold; that is, the word on the right can occur alone. We can conclude that the elements in brackets are optional dependents. Considering this dependency, we may suggest that the possessor in (32b) and the complex predeterminer in (32c) are also part of an extended projection, similar to the PP argument of the adjective in (32a). Schematically, we can illustrate this such that the bracketed elements in (32) are in Spec,XP, and the elements on the right form the head X:

¹⁶ Note that *alle die Väter* 'all the fathers' (without the complex adjectival element; cf. (32a)) is slightly marked (for discussion and references, see Roehrs to appear, chapter 2, section 6).



Note that XP itself is in a specifier position in the extended projection line of the head noun. Having said that, none of the dependents in Spec,XP involve (possessive) genitive. Given Spec-head agreement, that means that X and thus XP do not have a (possessive) genitive feature on them. As such, they do not raise issues for the agreement chain.¹⁷ Having discussed Spec-head agreement as one component of the proposal, I turn to selection in the next subsection.

3.2. *Fleshing out the Proposal*

Let us consider how the basic proposal works adding some more details in the process. I start with cases that do not involve a left periphery:

- (34)
- a. sein-e Freunde
his-INFL friends
'his friends'
 - b. Peters Freunde
Peter's friends
'Peter's friends'

These cases project a regular DP structure, and their derivation proceeds without problems (for some more detailed discussion, see, among many others, Alexiadou *et al* 2007, Julien 2005, Roehrs 2009). Next, we add elements in the left periphery, but the DP-level does not involve a Saxon Genitive:

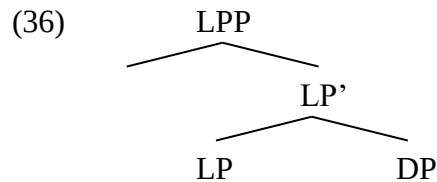
- (35)
- a. all-e sein-e Freunde
all-INFL his-INFL friends
'all his friends'
 - b. aus Jena sein-e Freunde
from Jena his-INFL friends
'his friends from Jena'

¹⁷ Similar to the extended projections of adjectives (Corver 1997), complex possessive constructions have been proposed to involve extended projections as well (Roehrs 2020). As to *ein* in (32c), it is interesting to note that this element cannot be reduced to 'n when it is preceded by *so manch* 'some':

- (i)
- a. 'N Mann kam.
a man came
'A man came.'
 - b. * So manch 'n Mann kam.
so some a man came

This might indicate that *ein* is not a regular article here offering some evidence for a more complex structure. Having said that, note that the account also goes through if the dative possessor and the complex predeterminer are not part of extended projections, but rather they are in Spec,DP. Either way, they do not involve (possessive) genitives and do not bring about issues for the agreement chain.

Arguing that *alle* ‘all’ and *aus Jena* ‘from Jena’ are in the LPP-level, this type of derivation succeeds without any problems as well – here the head LF selects an “agreeing” DP:



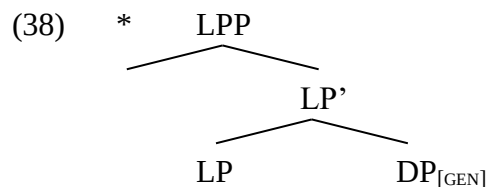
Briefly commenting on the possessive determiners in (35), in the older Germanic languages, possessives like *sein* ‘his’ involved genitive pronouns. Over time, these elements were reinterpreted as possessive determiners exhibiting agreement inflection in concord with the head noun (see Braune & Reiffenstein 2004: 245). As such, I assume that they do not involve genitive features. So far, so good.

Turning to the presence of (possessive) genitives, there are three options as regards such genitives in prenominal positions: only a Saxon Genitive is present, only a genitive (full) DP occurs, or a Saxon Genitive and a genitive DP co-occur.

As just discussed, Saxon Genitives can appear in prenominal position. Adding a PP in postnominal position is fine too (37a) – no LPP is projected. As demonstrated in Section 2.1, this is different when the PP is preposed, and LPP is present (37b):

- (37)
- a. Peters Anruf aus Jena
Peter’s call from Jena
‘Peter’s call from Jena’
 - b. * aus Jena Peters Anruf
from Jena Peter’s call

Above, I observed that if the left periphery is present, the constituent below it must involve an “agreeing” DP. A simple and straightforward way to capture this restriction involves selection: as concisely formulated in Chomsky (2000: 133), the selectee (the complement) must satisfy the selectional requirements of the selector (the head). I propose that LP must select an “agreeing” DP. On this assumption, we can propose that a DP with a (possessive) genitive feature on it is not an “agreeing” DP and, consequently, it does not satisfy the selectional requirements of LP:



Let us make the notion of “agreeing” more concrete.

I point out again that there are two types of genitives for current purposes. On the one hand, we find cases involving regular concord where genitive is assigned by a higher head and is shared by all relevant elements in the projection line of the head noun (39a). On the other hand, there are cases involving possessives where genitive is assigned within the noun phrase

(presumably by the possessum noun), but this case feature is not shared by the other elements of the noun phrase (39b):¹⁸

- (39) a. trotz [mein-es gut-en Freund-s]
 despite my-GEN good-GEN friend-GEN
 ‘despite my good friend’
 b. [Peters] gut-er Freund
 Peter’s good-NOM friend.NOM
 ‘Peter’s good friend’

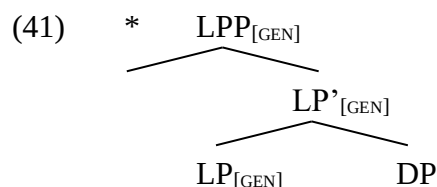
The difference between these two cases is the origin of the genitive and what it is assigned to: it is assigned outside of the DP in (39a) and to the whole DP but inside of it in (39b) and to a subpart of the DP.

As is often assumed, genitive assignment inside the noun phrase is related to a possessive relation (defined in broad terms). As such, a simple way to distinguish the two genitives is by the feature [$\pm$ POSS]. We can state then that LP is incompatible with a DP marked for [$+$ POSS] genitive. Note now that all elements in LPP discussed so far involve items not marked for (possessive) genitive. Thus, given Spec-head agreement, LP itself is [$-$ POSS genitive]. I suggest that this marking on LP results in an incompatibility with a [$+$ POSS genitive] feature on the DP. I propose that it is this incompatibility that leads to ungrammaticality.

As a second option, we could expect instances where the (possessive) genitive is only on LPP. These cases involve fronted genitive DPs. However, such cases are ungrammatical as well comparing (40a) to (40b-c):

- (40) a. seine / eine Eroberung der Gallier
 his-INFL/ a-INFL conquest the.GEN Gauls
 ‘his/a conquest of the Gauls’
 b. * der Gallier seine Eroberung
 the.GEN Gauls his-INFL conquest
 c. ?* der Gallier eine Eroberung
 the.GEN Gauls a-INFL conquest

With no Saxon Genitive present, the selectional requirements of LP are met. This means that we have to find a different way to rule out the following constellation:¹⁹



¹⁸ I leave open here whether *Peter* or *Peters* is the genitive form (for discussion, see Roehrs & Sprouse in prep. and references cited therein).

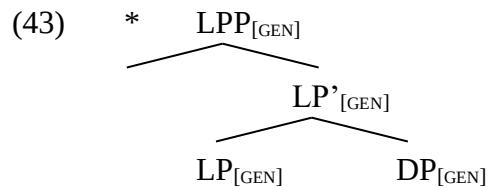
¹⁹ Note that this also involves a [$+$ POSS genitive] feature and a [$-$ POSS genitive] feature in a local domain – just switched around as compared to the option just discussed above. One might claim that this also leads to incompatibility. While this may be the case, the third option to be discussed below will show that this cannot be the explanation for all genitive DPs fronted to LPP.

Note that the genitive phrase *der Gallier* ‘the Gauls’ is a DP that is specified for definiteness. As such, I propose that it has to move through Spec,DP to check the definiteness feature on D. However, the determiner elements in (40) have already checked the definiteness feature. I propose that the genitive phrase cannot skip this potential landing site on its way to LPP ruling out this derivation.²⁰ Note that fronted PPs do not have to move through Spec,DP as they have nothing to do with definiteness. They can move to LPP in one fell swoop (for details, see Roehrs 2020).

The third option involves two (possessive) genitives. To begin, such cases are possible if a Saxon Genitive is prenominal and a genitive DP is postnominal (42a). However, similar to the second option, the postnominal genitive cannot be fronted (42b):

- (42) a. Cäsars Eroberung der Gallier
 Caesar’s conquest the.GEN Gauls
 ‘Caesar’s conquest of the Gauls’
 b. * der Gallier Cäsars Eroberung
 the.GEN Gauls Caesar’s conquest

With a genitive DP moved to LPP, there are two genitive features now, one on LPP and one on DP. Technically, this does not lead to a problem as the LP head selects a DP that has the same feature on it. However, similar to the second option, the genitive DP crosses over DP on its way to LPP skipping a potential landing site. Like the second option, I assume that this leads to ungrammaticality:



Taking stock, the only derivation that succeeds is the one that does not involve a (possessive) genitive, neither on the DP nor on the LPP (or both). Selection and Spec-head agreement rule out Saxon Genitives as heads of the agreement chain in German – LP must select an “agreeing” DP (general constraints on movement ban the cases involving genitive features on LPP). Crucially, the phenomenon in question only manifests itself in the DP-layer and only if the left periphery is filled. Furthermore, the account has to do with the transition from DP to LPP. As seen in the introduction, concord may hold between all elements in the noun phrase, from LPP down to NP. The only exceptions are PPs and Saxon Genitives (and dependent elements such as degree words, arguments of adjectives, etc. lower in the structure). The above account demonstrates that there is an additional requirement imposed on the features of DP when LPP is present.

²⁰ This can be made more formal by employing the Minimal Link Condition (e.g., Rizzi 1990, Chomsky 1995), something I will not do here.

3.3. Some Extensions of the Proposal

There are more cases falling under the Generalization of the Left Edge. First, like *alle*, *diese* ‘these’ can also precede a DP:

- (44) a. dies-e sein-e Freunde
these-INFL his-INFL friends
‘these friends of his’
b. * sein-e dies-e Freunde
this-INFL these-INFL friends

In fact, given some ordering restrictions, both *alle* and *diese* can precede a DP:

- (45) a. all-e dies-e sein-e Freunde
all-INFL these-INFL his-INFL friends
‘all these friends of his’
b. ?? dies-e all-e sein-e Freunde
these-INFL all-INFL his-INFL friends
c. * dies-e sein-e all-e Freunde
these-INFL his-INFL all-INFL friends

Note that uninflected *all* exhibits basically the same judgments.

As in the cases above, if the possessive determiner is replaced by a Saxon Genitive, ungrammaticality results (for (46a), see also Demske 2001: 248 fn. 19; for (46b), see also Bhatt 1990: 218):²¹

- (46) a. ?* dies-e Peters Freunde
these-INFL Peter’s friends
b. * all-e dies-e Peters Freunde
all-INFL these-INFL Peter’s friends

Again, uninflected *all* exhibits the same basic judgments. I propose that similar to *all(e)*, *diese* is in LPP when it precedes a possessive. Consequently, the account above carries over to these cases. Note that if this indeed falls under the same generalization and account, then it implies that there is a selecting head LP; that is, *diese* is not adjoined to DP in these cases. There is another type of extension.

I observed in Section 3.1 that there are no visible effects of the current phenomenon in the transition from the nominal to a higher domain, namely when a DP is selected by a verb, a preposition, or an adjective. I now turn to the discussion of a DP selected by a noun.

²¹ Note that the examples improve with a pause between *diese* and *Peters*, which presumably involves an appositive construction. Also, in order to explain the contrast between (44a) and (45a) vs. (46a-b), Bhatt (1990: 217-18) claims that *alle*, *diese*, and *seine* are heads. She proposes that *alle* is adjoined to *diese* and the resulting complex is adjoined to *seine*. The entire complex is inside Spec,DP. Bhatt claims that the three determiners can undergo head-to-head agreement. This is not possible with Saxon Genitives in (46) as the latter do not have inflections. Note though that head-to-head agreement is not a commonly accepted operation and that (complex) heads are typically not assumed to be in specifier positions.

It appears that the difference between possessive determiners and Saxon Genitives also holds in the transition from DP to a higher N. Consider (47a-b) (the data are from Haider 1992: 330):

- (47) a. die Eröffnung sein-es Geschäft-s
the opening his-GEN store-GEN
'the opening of his store'
b. * die Eröffnung Peters Geschäft-s
the opening Peter's store-GEN
'the opening of Peter's store'

However, one might object that this has to do with the so-called Genitive Rule in German. Specifically, note that the examples are also ungrammatical if the possessives are missing, both in the singular and plural:

- (48) a. * die Eröffnung Geschäft-s
the opening store-GEN
b. * die Eröffnung Geschäfte
the opening stores.GEN

One might argue on independent grounds that the example involving a singular count noun in (48a) is ungrammatical because a determiner is missing. However, this is different for the instance with a plural noun in (48b), which does not require a determiner. That these cases are also ungrammatical in the plural indicates that nouns in the genitive in German are subject to an additional, independent requirement, namely genitive nouns must be preceded by an element that displays a genitive ending in concord with it (for detailed discussion of the Genitive Rule, see Gallmann 1997):

- (49) a. der Genuss kalt-en Bier-es
the enjoyment cold-GEN beer-GEN
'the enjoyment of cold beer'
b. die Eröffnung neu-er Geschäfte
the opening new-GEN stores.GEN
'the opening of new stores'

Now, if the objection above were justified, then we would expect that the addition of an (inflected) adjective would make (47b) above perfect. This, however, is not the case ((50b) is from Gallmann 1997: 78):²²

²² Note that the example in (50b) is not fully ungrammatical for me; in other words, the addition of the adjective seems to make the example a bit better. Interestingly, other authors provide different judgments; for instance, Schachtel (1989: 106) finds the following example grammatical:

(i) die Gewalt Normans feinsinnig-er Argumente
the power Norman's sophisticated-GEN arguments
'the power of Norman's sophisticated arguments'

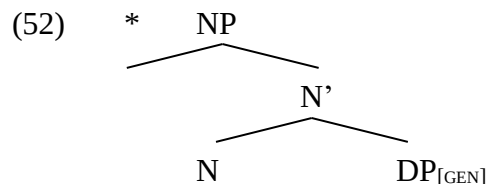
I am not sure if this involves dialectal variation. As far as I know, such differences in judgments are absent with the cases involving the left periphery. I continue the discussion with the ungrammatical case in the main text.

- (50) a. das Bild ihr-es verstorben-en Mann-es
 the picture her-GEN deceased-GEN husband-GEN
 ‘the picture of her deceased husband’
 b. * das Bild Frau Müllers verstorben-en Mann-es
 the picture Mrs. Müller’s deceased-GEN husband-GEN
 ‘the picture of Mrs. Müller’s deceased husband’

Observe that the Genitive Rule is not violated in (50b) – an element with a (concord) genitive ending precedes the head noun. Still, the example is ungrammatical. Before proceeding, note that this does not seem to have to do with genitives in general: cases where genitive is assigned by a non-nominal head fare better (notice that genitive verbs as in (51c) are rare and are typically used in formal contexts):²³

- (51) a. trotz Peters Abwesenheit
 despite Peter’s absence
 ‘despite Peter’s absence’
 b. Wir sind uns Peters Unterstützung gewiss.
 we are REFL Peter’s support certain
 ‘We are certain of Peter’s support.’
 c. ? Wir gedenken Peters Tod-es.
 We commemorate Peter’s death-GEN
 ‘We commemorate Peter’s death.’

If these considerations are on the right track, then there is a second type of case where Saxon Genitives cannot appear in the DP-layer, namely if a “non-agreeing” DP is selected by a higher head noun:



I propose that like the LP head, N cannot select a DP specified as [+POSS genitive]. Note now that LP and N are both nominal in nature: LP is part of the projection line of the lower head noun; N involves a higher head noun. While I currently have no interesting way to make this more formal, I tentatively suggest that this nominal property of LP and N is incompatible with a “non-agreeing” DP.

3.4. Some Consequences and Predictions of the Proposal

There are some interesting consequences of this proposal. While this paper focuses on German, the current analysis straightforwardly explains the main set of cross-linguistic data under investigation in Cirillo (2016). We saw above that Saxon Genitives cannot intervene between

²³ It is worth pointing out that the Genitive Rule is usually discussed in the context of complex DPs (where nouns take DP complements in the genitive).

all(e) and the lower part of the nominal in German. As noted by Cirillo (2016), this is different for English and Dutch where they can (37a-b). We can add Yiddish to these two languages, which works in a similar way (37c):

- (53) a. all John's friends
 b. al Jans vrienden (Dutch)
 all Jan's friends
 'all John's friends'
 c. oyf ale Peretses forlezungen in Varshe (Yiddish)
 at all Perets' presentations in Warsaw
 'at all Perets' presentations in Warsaw'
 (from *Forverts* 11/27/2009)

Since English, Dutch, and Yiddish do not have (morphological) genitive case, LP can easily select DP explaining the grammaticality of these examples.²⁴ If this is so, then there are four types of predictions: (i) only languages that have genitive case should exhibit the restriction discussed above, (ii) if LP is indeed responsible for this restriction, then languages that allow possessives to occur in lower prenominal positions should tolerate Saxon Genitives in these lower positions as well, (iii) if LP is absent as in adjunction structures, then Saxon Genitives in the DP-layer should be grammatical, and (iv) if a Saxon Genitive is in Spec,DP but more deeply embedded, then the restriction should not surface. As we will see, all four predictions are borne out.

Starting with the first prediction, Cirillo (2016: 215) states that the analysis of the Scandinavian languages should be fruitful. He provides the following basic data pointing out that the Icelandic example in (54b) sounds antiquated:

- (54) a. alla Johans vänner (Swedish)
 all Johan's friends
 'all Johan's friends'
 b. ? allir Jóns vinir (Icelandic)
 all John's friends
 'all John's friends'

He does not go into any details. Here, I will examine these and other cases more closely.

As can be seen in (55a-b), Swedish – similar to (54a) above – and Norwegian allow a Saxon Genitive when the left periphery is occupied ((55a-b) are from Vangsnes 1999: 125 fn. 21, 126):

²⁴ Cirillo (2016: 180) states that German has a true genitive but that English and Dutch assign genitive by *of* and *van* (note that Cirillo claims that English still has the vestige of a true genitive case in instances like *friends of John's*, p. 190). The ending -s in the latter two languages is interpreted as a possessive adjective that moves from Poss, the head of an intermediate PossP, to D. The various distinctions of the three languages are derived by a complex interplay of the different assumptions. It would take me too far afield to discuss this in detail here. Note though that I pointed out in Section 2.3 that there are some issues in Cirillo's account of German involving preposed PPs and uninflected *all*. Having said that, I think that Cirillo's basic claim that German has true genitive while English and Dutch do not is on the right track, and I will explore this in more detail below.

- (55) a. för alla Sveriges herrgårdsvagnar (Swedish)
 for all Sweden's station.wagons
 'for all Sweden's station wagons'
 b. alle Peters uangripelige teorier (Norwegian)
 all Peter's untouchable theories
 'all Peter's untouchable theories'

This is different for Icelandic. Note first that possessives are usually in postnominal position in this language (e.g., Sigurðsson 1993, 2006; Thráinsson 2007: 117ff). However, they can occur in prenominal position with certain restrictions; for instance, they must involve one-word elements (Pfaff 2015: 162). Relevant for current purposes, Sigurðsson (1993: 181) provides data that shows that Saxon Genitives cannot occur if the left periphery is occupied:

- (56) a. allar þínar þrjár nýju kenningar (Icelandic)
 all your three new theories
 'all your three new theories'
 b. * allar Jóns þrjár nýju kenningar
 all John's three new theories
 'all John's three new theories'

There is a second context in Icelandic, where the restriction could potentially surface.

Pfaff (2015, 2017) discusses noun phrases of the pattern “Adj-ST Det N”, his Pattern IV, where a strong adjective appears to the left of determiners like free-standing articles or demonstratives. He proposes that the strong adjective is in the left periphery of the noun phrase, similar to the cases involving *allir* ‘all’ above. If so, then we can test the different kinds of possessives in the DP-layer in this type of structure.

As pointed out by Pfaff (2015: 83, 2017: 311), a possessive determiner is fine (57a) (note that he does not provide data involving Saxon Genitives). Halldór Sigurðsson (p.c.) points out to me that (57a) is also fine for him but that (57b) is marked, either as ?? or as * (stating that he would generally prefer the possessive in postnominal position):²⁵

- (57) a. flest-ar mínar hugmyndir (Icelandic)
 most-ST my ideas
 'most of my ideas'

²⁵ Thráinsson (2007: 119) provides facts similar to (56a-b) above. However, he also points out that possessive pronouns in the genitive are fine in the context of *allir* ‘all’ (ia). Halldór Sigurðsson (p.c.) observes that possessive pronouns in the genitive are fine in Pattern IV as well (ib):

- (i) a. allar hennar þrjár nýju kenningar (Icelandic)
 all her.GEN three new theories
 'all her three new theories'
 b. flest-ar hans hugmyndir
 most-ST his ideas
 'most of his ideas'

Assuming that this involves a regular genitive feature, this must mean that this feature does not spread to the larger DP. Currently, I have no interesting proposal to account for the grammaticality of (ia-b).

- b. ??/* flest-ar Jóns hugmyndir
 most-ST John's ideas
 'most of John's ideas'

In other words, the different types of possessives pattern in the same way in Pfaff's Pattern IV as they do in the cases involving *allir* 'all' above.

As is well known, Swedish and Norwegian do not have genitive case, but Icelandic does. As just documented, the former languages allow a Saxon Genitive if the left periphery is occupied, but the latter does not. Thus, like the West Germanic languages, the presence or absence of genitive in these languages seem to correlate with the presence or absence of the restriction on Saxon Genitives in prenominal position. If this turns out to be true of all Germanic languages, then the head LP seems to have the same properties across this language family. At this point, I have not investigated at all if this goes beyond the Germanic languages.

As to the second prediction, older varieties of German allow possessives to occur in lower prenominal positions. This is illustrated with a possessive pronominal in Old High German (OHG) in (58a). If LP is indeed responsible for the effect, then we expect that Saxon Genitives should be fine in a low prenominal position. This is the case as shown in (58b) ((58a) is taken from Harbert 2007: 155, (58b) is taken from Demske 2001: 227):

- (58) a. in dheru sineru heilegun chiburdi (OHG)
 in the his holy birth
 'in his holy birth'
 b. in dhemu heilegin daniheles chiscribe
 in the holy Daniel's scripture
 'in Daniel's holy scripture'

Early New High German (ENHG) exhibits similar facts, here illustrated with a possessive pronominal and a genitive DP (data are taken from Demske 2001: 227):

- (59) a. andere jhre beschwerten (ENHG)
 other their complaints
 'their other complaints'
 b. vielfeltiges de-s Graff-en bitten
 manifold the-GEN count-GEN requesting
 'manifold requesting of the count'

It is undebated that these older varieties of German have genitive case, just like Modern German. Again, these facts are explained if LP poses a restriction on its complement (but other functional heads lower in the structure do not).

Turning to the third prediction, I proposed above that LP can only select an "agreeing" DP. Consider now floating quantifiers involving *alle* 'all' (60a). Note that (60a) involves a Saxon Genitive in the DP-layer of the DP related to the lower quantifier. It is usually argued that both elements, the DP and the related quantifier, originate together forming a constituent and that the DP undergoes further movement (e.g., Sportiche 1988). Given current assumptions, this cannot involve *alle* in LPP, which would involve LP selecting a "non-agreeing" DP in (60a) in underlying position. Interestingly, Bošković (2004) proposes that the floating quantifier is

merged as left adjunction to DP (60b), and the lower DP can move up stranding the quantifier in its position in (60a):

- (60) a. Peters Freunde sind alle gekommen.
 Peter's friends are all-INFL come
 'Peter's friends have all come.'
 b. [_{DP} alle [_{DP} Peters [_{NP} Freunde]]]

Since adjunction does not involve a selecting head, the base structure in (60b) is fine and the derivation can proceed successfully yielding (60a).

There seems to be independent evidence that floating quantifiers involve adjunction in German. Note that this language allows complex quantifiers like *alle drei* 'all three' and *alle beide* 'all (both)' to be stranded:

- (61) a. Die Männer sind alle drei verschwunden.
 the men are all three disappeared
 'All the three men have disappeared.'
 b. Wir sind alle beide unglücklich.
 we are all both unhappy
 'Both of us are unhappy.'

Crucially, if not stranded, these complex quantifiers do not form felicitous strings with the following DP:

- (62) a. * alle drei die Männer
 all three the men
 b. * alle beide wir
 all both we

This means that these quantifiers do not form regular nominal structures with the following DP. Rather, they seem to be binominals involving adjunction, where the DP (for some reason) has to move up the structure stranding the complex quantifier. Consider another, possibly related case.

As just stated, adjunction does not involve the head LP. This idea might be extendable to another instance pointed out by Haider (1988: 53):

- (63) a. fast sein ganz-es Einkommen
 almost his entire-INFL income
 'almost his entire income'
 b. ? fast Peters ganz-es Einkommen
 almost Peter's entire-INFL income
 'almost Peter's entire income'

Similar to above, the modifier *fast* 'almost' is in the left periphery. Notice though that the grammaticality contrast between (63a) and (63b) is less sharp than in the cases discussed in the first part of the paper. Similar to the floating quantifiers just discussed, this could imply that the

modifier *fast* is not in LPP but has a different analysis. Note in this regard that Haider (1988) suggests that *fast* is adjoined to DP. I turn to the fourth and final prediction.

Perhaps unexpectedly, the aforementioned judgments are reversed in coordination structures such that (non-agreeing) Saxon Genitives are fine but (agreeing) articles are not (data taken from Bhatt 1990: 142):

- (64) a. all-e mein-e und Peters Bücher
all-INFL my-INFL and Peter's books
all mine and Peter's books'
b. * all-e mein-e und di-e Bücher
all-INFL my-INFL and the-INFL books

The data find an explanation in the current context by assuming that only the first conjunct is relevant to the agreement chain; in other words, given certain assumptions, the second conjunct involving the Saxon Genitive is too deeply embedded to be relevant.

In more detail, there are different ways to analyze these coordination structures. Here, I assume that the coordination involves the two possessives indicated in (65a) by square brackets. Furthermore, I assume that the conjunction head *und* 'and' projects a regular X-bar structure such that the first conjunct is in its specifier position and the second conjunct is in its complement position (65b). This yields a Conjunction Phrase (&P; see Munn 1987 and Kayne 1994: 59, but cf. also Munn 1992, 1993):

- (65) a. alle [meine] und [Peters] Bücher
b. [&P [Conjunct 1] [&' and [Conjunct 2]]]

Note now that the Saxon Genitive is in the complement position of &P. Given this, its (possessive) genitive feature cannot spread to &P by Spec-head agreement and consequently, is not on D or DP.²⁶ This explains the grammaticality with Saxon Genitives.²⁷

As for the ungrammaticality of (64b) above, note first that if the article is replaced by a demonstrative, the example is fine (66a). Notice that if the article is stressed, indicated below by capitalization, it functions as a demonstrative in German and the examples are grammatical too (66b-c).²⁸

- (66) a. all-e mein-e und dies-e Bücher
all-INFL my-INFL and these-INFL books
'all mine and these books'

²⁶ If the coordination structure involves different types of conjuncts, then we can state that no relevant chain is established with the Saxon Genitive in the second conjunct – there is simply no element in the left periphery of the second conjunct.

²⁷ For some unclear reason, coordination with the Saxon Genitive in the first conjunct is not fully ungrammatical:

(i) ?? all-e Peters und mein-e Bücher
all-INFL Peter's and my-INFL books
'all Peter's and my books'

²⁸ Unstressed *die* does not occur by itself (this presumably has to do with the fact that articles are often claimed to be clitics). In other words, *die* in (66c) must involve a demonstrative.

- b. all-e mein-e und DI-E Bücher (da)
all-INFL my-INFL and those-NOM books (there)
'all mine and these books (there)'
- c. DI-E and all-e mein-e Bücher
those-NOM and all-INFL my-INFL books
'these and all my books'

There are two ways to explain the ungrammaticality of (64b) above. First, note that the definite article in (64b) seems to “pick out” the referent made salient by *meine* in the first conjunct; that is, the two conjuncts refer to the same set of (referential) entities. This presumably leads to a semantically ill-formed coordination. This is different for demonstratives, which may refer to a different set of books. Second, considering that demonstratives involve stress and that articles are often assumed to be clitics, one could state that both conjuncts are quite different in their morpho-phonological makeup leading to ungrammaticality. Whichever option turns out to be true, there is presumably an independent explanation for the ungrammaticality of (64b).

To sum up, as some of the predictions above have made clear, the formation of an agreement chain is sensitive to structure: the restriction only holds in the left edge of the noun phrase, the left periphery and the DP-layer, and it is sensitive to the degree of embedding of the relevant elements.

4. Further Extensions

First, I turn to the discussion of some other structural points, specifically, the question of where certain quantifiers can appear in the noun phrase. This includes a discussion of some less widely known facts in Yiddish. In the second part, I discuss an additional adjacency requirement in Possessive Doubling constructions.

4.1. Structural Components of the Proposal

First, I turn to some structural considerations arguing that not all seemingly similar cases follow from this additional requirement for agreement chains but may have an independent explanation. In this regard, I consider the possible positions of *alle* ‘all’ and *beide* ‘both’. Let us return to the data from the introduction:

- (67) a. * all-e Marias Bücher
all-INFL Mary’s books
'all Mary’s books'
- b. * beid-e Marias Bücher
both-INFL Mary’s books
'both Mary’s books'

Taking the data at face value, one could be inclined to give them the same analysis. In fact, Bayer (2015) tentatively suggests just that, namely that the Saxon Genitive disrupts an agreement chain in (67b) just as it does in (67a). However, as also pointed out in the introduction, *beide* does not behave like *alle* in all other respects:

- (68) a. * Marias all-e Bücher
 Mary's all-INFL books
 b. Marias beid-e Bücher
 Mary's both-INFL books
 'both Mary's books'

In what follows, I will argue that (67b) is independently out, and its ungrammaticality has nothing to do with an ill-formed agreement chain.

Let us consider the positional options of *alle* and *beide* in more detail. Here I provide English examples again. Their contrast to German will make the point under discussion clearer. Starting with English, *all* must precede determiners (69) to the extent they are present (70):

- (69) a. all his books
 b. all these books

- (70) a. all books
 b. * the all books

The same facts hold for German:

- (71) a. all(-e) sein-e Bücher
 all-INFL his-INFL books
 'all his books'
 b. all(-e) dies-e Bücher
 all-INFL these-INFL books
 'all these books'
- (72) a. all*(-e) Bücher
 all-INFL books
 'all books'
 b. * di-e all-e(-n) Bücher
 the-INFL all-INFL.ST(/-.WK) books

It is interesting to note that the inflection on German *alle* is optional when a possessive determiner or demonstrative is present (71a-b) but obligatory when a determiner is absent (72a).²⁹ Given this difference in inflections in German, let us assume that there are two different positions for English *all* and German *all(e)*:

²⁹ Recall from Footnote 16 that unlike *all die Väter* 'all the fathers', inflected *alle* occurring before *die Väter* leads to slight markedness. Interestingly, if a preposed PP is added to such cases, both examples become worse:

- (i) a. (?) von Peter all di-e Autos
 of Peter all the-INFL cars
 'all the cars of Peter's'
 b. ?? von Peter alle di-e Autos
 of Peter all-INFL the-INFL cars
 'all the cars of Peter's'

- (73) LPP DP
 all *all* (English)
 all(e) *alle* (German)

Thus far in the discussion, English and German are basically the same. However, these two languages differ as regards *both* and *beide*.

Similar to *all*, English *both* can occur before a determiner (74a-b), it can be the determiner itself (54a), but it cannot follow the determiner (75b):³⁰

- (74) a. both his friends
 b. both these friends

- (75) a. both books
 b. * the both books

This is different for German *beide*. Notice that *beide* cannot precede a determiner (weak endings on the possessive or the demonstrative make the examples worse).³¹

- (76) a. ?? beid-e sein-e(?-n) Bücher
 both-INFL his-INFL.ST(/-.WK) books
 b. ?? beid-e dies-e(?-n) Bücher
 both-INFL these-INFL.ST(/-.WK) books

However, *beide* can easily follow a determiner:

- (77) a. sein-e beid-en Bücher
 his-INFL both-INFL.WK books
 ‘both his books’
 b. dies-e beid-en Bücher
 these-INFL both-INFL.WK books
 ‘both these books’

These two data points make German *beide* different from English *both*. As perhaps expected, German *beide* can also be a determiner. Like other determiners, it triggers a weak ending on a following adjective (78a-b). Furthermore, we already observed that *beide* has to follow a determiner if there is one. In fact, if *beide* follows a determiner, it behaves like a regular adjective in that it takes a weak ending (78c):

³⁰ There is a data point gleaned from the TV sitcom *Everybody loves Raymond*, which has *both* following the definite article:

(i) the both of you

Crucially though, there is no overt noun after *both*. This shows that ellipsis contexts allow other possibilities implying a different syntax.

³¹ Vater (1991:18) judges *beide diese Bücher* as fine. However, I side with Bhatt (1990: 217), who finds these types of example degraded.

- (78) a. beid-e klein-en Bücher
 both-INFL small-INFL.WK books
 ‘both small books’
 b. ?? beid-e klein-e Bücher
 both-INFL small-INFL.ST books
 c. di-e beid-en klein-en Bücher
 the-INFL both-INFL.WK small-INFL.WK books
 ‘both the small books’

We conclude that besides a determiner, *beide* can also be an adjectival element in German.

To sum up thus far, there are three positions for *both/beide*. Consider (79). Note that both languages share the option of having this element in the DP-layer, but they differ in the second option in that English *both* can occur above the DP-level but German *beide* below it. I take the latter element to be an adjectival quantifier in CardP. To be clear then, none of the two languages tolerates this element in all three positional options (but as documented below, *ale* in Yiddish can appear in all three):

- (79) LPP DP CardP
both both (English)
beide beiden (German)

Comparing (69) to (78), English *all* and *both* have the same distributional options but German *all(e)* and *beide* do not. Continuing with German, we may state that if *alle* or *beide* is present, they surface in the DP-level. If a(nother) determiner is present, it forces the quantifier to be in a higher (*alle*) or lower (*beide*) position. Presumably, both elements share the option of surfacing in the DP-layer as definite DPs require an overt element in the DP-level (Longobardi 1994, Julien 2005, and Cirillo 2016). Thus, besides the restriction discussed above, the DP-layer also plays an important part in the distribution of determiner(-like) elements in definite DPs in German.

The different positions of *alle* and *beide* in German are confirmed by other facts. In this language, both elements can co-occur provided *beide* is the adjectival version (the weak ending -*n* added to *beide* in (80a) makes the example perfect):³²

- (80) a. all-e beid-e²(-n) Bücher
 all-INFL both-INFL.ST(/-.WK) books
 ‘both (of the two) books’
 b. * beid-e all-e(-n) Bücher
 both-INFL all-INFL.ST(/-.WK) books

English disallows both elements to occur at the same time:

³² Without an overt noun, Duden (2016: 310) provides *alle beide* (also in *Datenbank für Gesprochenes Deutsch* [DGD; Database of Spoken German], often occurring as a complex floating quantifier); DGD also shows *alle beiden* (some of these instances have an overt noun). Given these inflectional options of *beid-*, the two analyses for (80a) are as follows:

- (i) a. ? [LPP alle [DP beide [NP Bücher]]]
 b. [DP alle [CardP beiden [NP Bücher]]]

Note though that if (ia) is possible, it is not clear why the following is bad: **all beide Bücher* (cf. *all diese Bücher*).

- (81) a. * all both books
 b. * both all books

Although rare, both elements in German can also co-occur with a determiner separating them:

- (82) a. all(-e) dies-e beid-en
 all(-INFL) these-INFL both-INFL.WK
 ‘all these (two)’
 b. all di-e beid-en
 all the-INFL both--INFL.WK
 ‘all the (two)’

Here are two examples retrieved by a *google* search (note that (83a) involves an incomplete sentence marked by three dots):

- (83) a. Ich habe kürzlich all diese beiden...
 I have recently all these both
 ‘Recently, I have ...all these (two)...’
 b. All diese beiden Elchbrüder wollen nur ihre Drachen hoch fliegen lassen
 all these both elk.brothers want only their kites high fly let
 ‘All these (two) guys only want to let their kites fly high.’

We are now ready to return to the data from the beginning of this subsection. With *Marias* in Spec,DP, it is clear that *beide Marias Bücher* is independently out as *beide* cannot appear in LPP in the first place. In other words, the ungrammaticality of this example is not an argument for an agreement chain and its disruption, an extension cautiously suggested by Bayer in his proposal.

The upshot of this discussion is that the possible structural positions of elements have to be taken into account in the discussion of agreement chains. In fact, it has become clear that cross-linguistically, quantifiers can be in different positions in different languages. Besides the DP-layer, they can also be above the DP-layer (English *both*; also Icelandic *margar* ‘many’, see Pfaff 2015: 84; and Old English *sum*, see Roehrs & Sapp 2018; more generally also Pfaff & Walkden 2024) or below the DP-layer (German *beide*; also Early New High German *alle* and Yiddish *ale*, see Roehrs & Sapp 2016: 144 and below). Note that the occurrence of quantifiers in the positions lower than LPP seems to be dependent on the inflection on the quantifier. This point can be made more explicit with Yiddish and German

Yiddish is interesting in that the universal quantifier can appear in all three positions (i.e., pre-DP, DP-layer, post-DP-layer), given certain restrictions. As observed by Jacobs (2005: 240), uninflected *al* occurs before a definite article, but inflected *ale* appears before a possessive article and in the DP-layer. As I will show below, there is another position that *ale* can occur in, not mentioned in Jacobs – this element can also occur between the DP-layer and numerals and adjectives; that is, *ale* can also surface lower in the nominal structure. To be clear, uninflected *al* only occurs if a definite article follows, but inflected *ale* can appear in all other positions. Given that Yiddish has not received much attention in this regard, let us consider these facts in more detail

Uninflected *al* can precede the definite article (84a). Interestingly, we find *al* and the definite determiner often spelled as one word (84b). The reduced form *alde* in (84c) may indicate a different status of this combination:

- (84) a. al di gute frages (Yiddish)
all the good questions
‘all the good questions’
(Jacobs 2005: 240)
- b. aldos dozike groyse beyz
all.the here great evil
‘all this great evil’
(from *Tanakh: Yirmeyohu*; Corpus of Modern Yiddish)
- c. Tsu al di / alde shvartse yor!
to all the / all.the black year
‘Darn it!’
(Lockwood 1995: 56)

That this special status of *al* may be true is shown by the fact that there are no examples of *al yene* ‘all those’ (for *al* and a possessive article, see Footnote 33 below). Indeed, Birnbaum (1979: 297, 256) provides another reduced form (*aldis*) and states explicitly that if *al* precedes the definite article, it is not inflected, and the two are felt to form a single word. He states that the two elements are joined together, which results in a weakened vowel.

As stated in Jacobs (2005), inflected *ale* can precede possessive determiners (85a). In addition, it can also precede demonstratives (85b). As expected, it can also occur without a determiner in the DP-layer (85c-d):

- (85) a. al-e mayn-e dray elter-e brider (Yiddish)
all-INFL my-INFL three older-INFL brothers
‘all my three older brothers’
(Jacobs 2005: 240)
- b. ale yene zakhn
all those things
‘all those things’
(U. Weinreich 1999: 192)
- c. al-e (dayn-e) gut-e frages
all-INFL your-INFL good-INFL questions
‘all (your) good questions’
(Jacobs 2005: 240)
- d. al-e bikher
all-INFL books
‘all books’
(Jacobs 2005: 187)

Perhaps most interestingly, this language also allows this quantifier to follow determiners. In this case, the quantifier is always inflected:

- (86) a. di ale shrayber (Yiddish)
 these all writers
 ‘all these writers’
 (Birnbaum 1979: 297)
- b. yene ale frages
 those all questions
 ‘all those questions’
 (Goldberg 1993: 46)
- c. di dozike ale zakhn
 these here-ADJ- all things
 ‘all these things’
 (Lockwood 1995: 56)
- d. dize ale morden
 these all murders
 ‘all these murders’
 (from *Di indishe printsesin*, p. 82)

As might be expected, inflected *ale* can also follow possessives:

- (87) a. zayne ale kleyder (Yiddish)
 his all clothes
 ‘all this clothes’
- b. Eriks ale verk
 Erik’s all works
 ‘all Erik’s works’
 (both from Lockwood 1995: 56)

Note that this *ale* occurs above numerals and adjectives:

- (88) a. di ale dray zakhn (Yiddish)
 the all three things
 ‘all the three things’
 (from *Keynemsdorf*, p. 64)
- b. di ale tayere zakhn
 the all expensive things
 ‘all the expensive things’
 (Olsvanger’s *Röyte pomerantsen*; story 150, page 97)

Finally, this *ale* can be modified by *kimat* ‘almost’ (89a), and it can occur after pronouns (89b-c) and as a floating quantifier (89d):

- (89) a. in di kimat ale lender (Yiddish)
 in the almost all countries
 ‘in almost all the countries’
 (from *Forverts* 11/12/2014)

- b. Ikh hob zey ale.
I have them all
'I have them all.'
(Katz 1987: 120)
- c. di ale, vos zey zenen geblibn
these all which they are stayed
'all those that stayed'
(Lockwood 1995: 124)
- d. mir zaynen shoyn ale geven in Tsfas
we are already all been in Safed
'we have all been to Safed already'

To summarize the empirical facts, in predeterminer position, we find *al* before definite articles and *ale* before demonstratives and possessive determiners.³³ In the DP-layer, we find inflected *ale*, and the same goes for the postdeterminer position. As far as I have been able to establish, *al* never occurs in the DP-layer or below, including cases when this quantifier is floated. Note that the distribution of this quantifier with the definite article seems to be special – it is the only context where it can be uninflected (and the article shows some signs of phonetic reduction). Interestingly, the uninflected form of this quantifier is also special in German.

Turning to German, the comparison between (90a) and (90b) shows that uninflected *all* can only occur if it precedes a determiner-like element. If no such element is present, this quantifier must be inflected (90c):³⁴

- (90) a. all di-e / dies-e / mein-e Bücher
all the-INFL / these-INFL / meine-INFL books
'all the / these / my books'
- b. * all Bücher
all books

³³ Note that exceptions seem to go in both directions. First, inflected *ale* can occasionally also occur before the definite article (although it is not impossible that *di* might be a demonstrative here):

- (i) ale di prekhlike bilder (Yiddish)
all the beautiful pictures
'all the beautiful pictures'
(from *Forverts* Radio Show 12/11/2011)

Second, in poetic language, one finds uninflected *al* before possessives (iia). In fact, I came across an example that uses an apostrophe to indicate the missing inflection (iib):

- (ii) a. al mayne blutn (Yiddish)
all my bloods
'all my bloods'
(googled: <http://www.cs.engr.uky.edu/~raphael/yiddish/nybc208866.html>)
- b. al' mayne treren-shtromen
all my tears-streams
'all my streams of tears'
(googled: https://github.com/pkarmstr/NYBC/blob/master/PyYiddish/mt-materials/cleaned_docs/nietzsche-zaratustra-2.txt)

³⁴ As mentioned above, inflected *alle* can occur before demonstratives and possessive determiners, but inflected *alle* is marked if it appears before definite articles as in *alle die Bücher* 'all the books'. As discussed in Roehrs (to appear), this might involve a phonological explanation.

- c. all-e Bücher
all-INFL books
'all books'

Recall that in Yiddish, uninflected *al* can only occur before the definite article; that is, its distribution is more restricted than in German.

I propose for German that both inflected *alle* and uninflected *all* can be merged in LPP and in the DP-layer alike. Inflected *alle* is not subject to any specific restrictions except that it has to merge in the DP-layer to license a definite DP if there is no other determiner element present, as in (90c) above. As for uninflected *all*, I propose that this is a deficient element that has to procliticize to a determiner element in German. If no such element is present, it leads to ungrammaticality (90b), independent of whether or not *all* merges in LPP or in the DP-layer. Yiddish is similar to German, except uninflected *al* has to procliticize to the definite article. To be clear then, this proposal explains the fact that only uninflected forms of the quantifier can occur in LPP, and it also explains the fact that only inflected forms of the quantifier can occur in the DP-layer and below including cases of quantifier float.

4.2. *Adjacency Requirement in Possessor Doubling Constructions*

As argued above, not all cases discussed this far can be explained by the disruption of an agreement chain in the sense of Bayer (2015). Rather, they all have to do with the appropriate head of a chain provided a chain is formed. As seen above, an abstract chain may start at the DP-level although the trigger of that chain, the selector, is in the left periphery.

There seems to be an additional requirement on the formation of this type of chain that becomes apparent when an element is added to the Possessor Doubling construction. Recall from the introduction that Bayer (2015) judges examples of type (91a) to be perfectly fine (with the qualification that he requires the presence of the proprial article before *Peter*). As already pointed out in Footnote 2, these are slightly marked for me. Interestingly, the possessor can also precede *alle* (91b):

- (91) a. ? all-e Peter sein-e Bücher
all-INFL Peter his-INFL books
'all Peter's book's'
b. (?) Peter all-e sein-e Bücher
Peter all-INFL his-INFL books
'all Peter's books'

While the judgments are very subtle, it seems that (91b) sounds a little better than (91a). When *alle* is replaced by uninflected *all*, the (a)-example gets worse:

- (92) a. ?? all Peter sein-e Bücher
all-INFL Peter his-INFL books
'all Peter's books'
b. (?) Peter all sein-e Bücher
Peter all his-INFL books
'all Peter's books'

Something similar holds for preposed PPs (93a) with the qualification that *Peter* cannot precede the PP (93b) (as above, the addition of the dative proprial article *dem* does not improve the example; for Haider 1992: 315, examples like (93a) are fully ungrammatical):

- (93) a. ?? nach Hamburg Peter sein Zug
 to Hamburg Peter his train
 ‘Peter’s train to Hamburg’
 b. * Peter nach Hamburg sein Zug
 Peter to Hamburg his train

That there is no general reordering between *Peter* and the leftmost element is confirmed by examples involving *diese*:

- (94) a. ? dies-e Peter sein-e Bücher
 these-INFL Peter his-INFL books
 ‘these books of Peter’s’
 b. * Peter dies-e sein-e Bücher
 Peter these-INFL his-INFL books
 ‘these books of Peter’s’

Starting with the (a)-examples in (91) to (94), we observe that if *Peter* intervenes between the trigger of the chain and the possessive determiner, the examples are marked. In fact, the examples are more marked if the trigger does not show any overt agreement (*all*, *nach Hamburg*). In other words, uninflected/non-agreeing elements in LPP (*all*, *nach Hamburg*) are worse than inflected/agreeing ones (*alle*, *diese*). As for the (b)-examples, the possessor *Peter* can only reorder with *all* and *alle* but not with *nach Hamburg* or *diese*. In fact, if *Peter* can precede the trigger of the chain as in the cases of *alle* and *all*, the examples seem to improve. In other words, to the extent possible, there seems to be a preference to have the two determiner(-like) elements, *all(e)* and the possessive determiner, adjacent.

Recall that preposed PPs are higher in the structure than *all(e)* but that *diese* is below this quantifier:

- (95) a. (?) von Peter all di-e Bücher
 of Peter all the-INFL books
 ‘all the books of Peter’s’
 b. all dies-e (mein-e) Freunde
 all these-INFL my-INFL friends
 ‘all these friends (of mine)’

I take this to mean that *Peter* in (93b) cannot move above the preposed PP for some independent reason. It is less clear though why *Peter* cannot move above *diese* in (94b). Specifically, with *diese* located below *all(e)*, *Peter* should be able to move above *diese* just as it can with *all(e)*. This, however, is not the case. We have seen in Section 2.1 that possessors of the Possessor Doubling construction are not subject to the restriction discussed above. Given this, it seems unlikely that an explanation for the slight grammaticality contrast can be found there. Thus, it is

not immediately clear how to make sense of this contrast involving *all(e)* as in (91b) and (92b) vs. *diese* as in (94b).

Bošković (2004: 723) claims that certain light elements are invisible to the Verb-Second constraint in German; that is, his analysis proposes a “delayed,” PF account of Verb-Second. Let us assume then that cases involving *all(e)* are special in that they also involve a “late” operation. Suppose that chains are best if they are not disrupted by a “non-agreeing” element, the possessor. In other words, if reordering is possible as in the cases of *alle* and *all*, then it can take place yielding an undisrupted agreement chain of two determiner-like elements.³⁵ That this is not possible with *diese* might suggest that this is a late, non-syntactic reordering. This seems to be confirmed by the observation in Cirillo (2016: 213) that the possessor *Peter* cannot move above *al* in Dutch (unlike German (92b)). I take this to mean that this late reordering is a language-specific, non-syntactic operation.

4. Conclusion

This paper has demonstrated that the empirical facts in the left part of the German noun phrase are fairly complex: there is a restriction such that when an element surfaces in the left periphery, the determiner(-like) element in the DP-layer has to agree with the lower part of the larger noun phrase. Following Olsen (1989a) and Bayer (2015), I have interpreted these cases as agreement chains. With concord brought about independently, the agreement chain discussed here is only built under certain conditions: the very presence of an element in the left periphery causes the abstract chain to exclude a (possessive) genitive feature. It was pointed out that the trigger may not coincide with the head of the chain. Indeed, if there is no element in the left periphery, no restriction manifests itself. Given this, it seems clear that the chain seems to work from the top down. This seems somewhat unusual for abstract chains where in syntax, (movement) chains are usually held to originate lower in the structure. To account for this unexpected directionality, I proposed that a higher head (LP) selects a DP that cannot have a (possessive) genitive feature. This, in effect, establishes a requirement for concord.

More generally, the question arises why this restriction only holds in the left periphery. One could state that the further away elements are from the head noun, the more restrictions seem to hold on them and their adjacent elements. One might suggest then that felicitous agreement chains facilitate parsing; that is, they signal what belongs to the lower nominal. The paper also discussed a number of extensions. For instance, it appears that only the Germanic languages with genitive case exhibit this restriction. It remains to be seen if this is a more general phenomenon cross-linguistically.

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³⁵ This leaves the degradedness involving a preposed PP in (93a) unaccounted for as there is only one determiner (i.e., *sein* ‘his’) present.

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